BRAC University

Department of Computer Science and Engineering

CSE422: Artificial Intelligence

Lab Project Report

Diabetes Prediction Using Machine Learning

Dataset	Diabetes Dataset (100,000 records)
Problem Type	Binary Classification
Models Used	KNN, Decision Tree, Logistic Regression, Neural Network, KMeans
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1. Introduction

Diabetes is one of the most prevalent chronic diseases globally, affecting hundreds of millions of people and placing an enormous burden on healthcare systems. Early detection is critical — it allows for timely medical intervention, lifestyle adjustments, and better long-term patient outcomes. Traditional diagnosis requires laboratory tests and clinical consultations, which may not always be accessible or timely.

This project aims to build and evaluate machine learning models that can predict whether a patient has diabetes based on a set of clinical and demographic features. The motivation is to demonstrate how data-driven approaches can supplement clinical decision-making, particularly in resource-limited settings where medical infrastructure is insufficient.

The project follows the complete machine learning workflow: data loading, exploratory data analysis, preprocessing, model training, and evaluation. Four supervised models — KNN, Decision Tree, Logistic Regression, and a Neural Network — are trained and compared. Additionally, KMeans clustering is applied to treat the problem as an unsupervised task and explore natural groupings within the data.

2. Dataset Description

2.1 Dataset Overview

The dataset used in this project is the Diabetes Prediction Dataset, which contains clinical and lifestyle information for 100,000 patients.

Property	Value
Number of Records	100,000
Number of Features	8 input features + 1 target
Target Variable	diabetes (0 = No Diabetes, 1 = Diabetes)
Problem Type	Binary Classification
Source	diabetes_dataset.csv

Why Classification? The target variable 'diabetes' is a discrete binary label (0 or 1), making this a classification problem. There is no continuous output to predict, ruling out regression.

Feature Types:

Feature	Type	Category	Description
age	Float	Quantitative	Patient age in years (0.08 – 80)
bmi	Float	Quantitative	Body Mass Index
HbA1c_level	Float	Quantitative	Glycated hemoglobin level

blood_glucose_level	Float	Quantitative	Blood glucose reading
hypertension	Integer (0/1)	Quantitative (Binary)	Has hypertension or not
heart_disease	Integer (0/1)	Quantitative (Binary)	Has heart disease or not
gender	String	Categorical	Female / Male / Other
smoking_history	String	Categorical	never/current/former/ ever/not current/No Info
diabetes	Integer (0/1)	Target	Diabetic (1) or Not (0)

Encoding Requirement: Yes. The features 'gender' and 'smoking_history' are categorical text variables that machine learning algorithms cannot process directly. They must be encoded into numerical form. Label Encoding is applied, converting each unique string value to an integer.

2.2 Correlation Analysis

A Pearson correlation analysis was performed on all features (after temporarily encoding categoricals). The results are shown below:

Feature	Correlation with Diabetes
blood_glucose_level	0.422
HbA1c_level	0.404
age	0.259
bmi	0.215
hypertension	0.196
heart_disease	0.163
smoking_history	0.097
gender	0.036
diabetes (self)	1.000

Observations from Correlation Analysis: blood_glucose_level and HbA1c_level have the strongest positive correlations with the target variable. This is clinically meaningful — both are primary diagnostic indicators for diabetes. age and bmi show moderate positive correlations (~0.21–0.26), consistent with known diabetes risk factors. hypertension and heart_disease show weak-to-moderate correlations (~0.16–0.20), suggesting comorbidities have some predictive value. gender and smoking_history have very weak correlations.

2.3 Imbalanced Dataset

The class distribution of the target variable 'diabetes' is highly imbalanced:

Class	Count	Percentage
0 (No Diabetes)	82,284	91.5%
1 (Diabetes)	7,634	8.5%
Total	89,918	100%

The dataset is highly imbalanced with a ratio of approximately 10.8:1 (No Diabetes : Diabetes). This means a naive model that always predicts 'No Diabetes' would achieve ~91.5% accuracy — which is misleading. Due to this imbalance, Stratified Splitting is used during train/test split to ensure both classes are proportionally represented. Recall (sensitivity) for the diabetic class is a more important metric than accuracy alone.

A bar chart of class distribution is included in the notebook (Section 3.2).

2.4 Exploratory Data Analysis

EDA was performed to extract important relationships from the data. Key findings:

- HbA1c_level and blood_glucose_level show clear bimodal distributions when split by diabetes class.
- Older patients (age > 50) have significantly higher rates of diabetes compared to younger patients.
- Higher BMI values correlate with increased diabetes prevalence, particularly BMI > 30.
- Patients with hypertension show a notably higher proportion of diabetes (~30%) compared to those without hypertension (~8%).
- Similarly, patients with heart disease show elevated diabetes rates (~20% vs ~8%).

3. Dataset Pre-processing

3.1 Fault: Missing / Null Values

Problem: The dataset contains missing values across all columns. The missing value counts before preprocessing are:

Feature	Missing Values
gender	9,835
age	10,065
hypertension	10,148
heart_disease	9,919
smoking_history	9,988
bmi	9,944
HbA1c_level	10,139
blood_glucose_level	9,880
diabetes	10,082

Solution: All rows containing any null/missing value were dropped (listwise deletion). This approach was chosen because missing values appear simultaneously across multiple columns in the same rows, making imputation unreliable. After dropping nulls, 38,636 clean rows remain.

3.2 Fault: Categorical Variables

Problem: gender and smoking_history are text-based categorical variables. Machine learning algorithms require numerical inputs.

Solution: Label Encoding was applied to both categorical columns. Each unique string value is mapped to an integer.

Feature	Label Encoding Mapping
gender	Female=0, Male=1, Other=2
smoking_history	No Info=0, current=1, ever=2, former=3, never=4, not current=5

3.3 Fault: Feature Scaling

Problem: Features have vastly different numerical ranges. For example, blood_glucose_level ranges from 80 to 300, while hypertension is binary (0 or 1). Models like KNN, Logistic Regression, and Neural Networks are highly sensitive to feature scale.

Solution: StandardScaler (Z-score normalization) was applied — transforming each feature to have zero mean and unit variance. Critically, the scaler was fit only on training data and then applied to test data to prevent data leakage.

4. Dataset Splitting

Stratified splitting was used due to the significant class imbalance (10.8:1 ratio). Stratification ensures both sets contain the same proportion of diabetic and non-diabetic patients.

Split	Samples	Percentage	Class Distribution
Training Set	30,908	80%	~2,615 diabetic / ~28,293 non-diabetic
Test Set	7,728	20%	654 diabetic / 7,074 non-diabetic
Total	38,636	100%	~3,269 diabetic / ~35,367 non-diabetic

5. Model Training & Testing

5.1 KNN Classifier

Configuration: n_neighbors=7. A distance-based classifier.

Class	Precision	Recall	F1-Score	Support
No Diabetes	0.96	0.99	0.98	7,074
Diabetes	0.89	0.57	0.70	654
Accuracy			0.9577	7,728

5.2 Decision Tree Classifier

Configuration: max_depth=8, random_state=42. The maximum depth is limited to 8 to prevent overfitting.

Class	Precision	Recall	F1-Score	Support
No Diabetes	0.97	1.00	0.98	7,074
Diabetes	0.96	0.67	0.79	654
Accuracy			0.9702	7,728

5.3 Logistic Regression

Configuration: max_iter=1000, random_state=42. A linear baseline classifier well-suited to binary classification.

Class	Precision	Recall	F1-Score	Support
No Diabetes	0.96	0.99	0.98	7,074
Diabetes	0.87	0.61	0.72	654
Accuracy			0.9590	7,728

5.4 Neural Network (MLPClassifier)

Architecture: Input(8) → Hidden Layer 1 (128 neurons) → Hidden Layer 2 (64 neurons) → Hidden Layer 3 (32 neurons) → Output (Sigmoid). Activation: ReLU. Early stopping was enabled.

Class	Precision	Recall	F1-Score	Support
No Diabetes	0.97	1.00	0.98	7,074
Diabetes	0.96	0.67	0.79	654
Accuracy			0.9696	7,728

5.5 KMeans Clustering (Unsupervised)

The dataset was treated as an unsupervised learning problem. KMeans clustering was applied to discover natural groupings in the data without using the target labels.

Elbow Method: The elbow method was used to determine the optimal number of clusters. Inertia was computed for k=2 through k=10. The elbow curve shows a significant bend at k=2, confirming that 2 clusters is the most natural grouping — consistent with the binary nature of the target.

KMeans with k=2 was fitted on the training data. PCA was used to reduce the 8-dimensional feature space to 2 dimensions for visualization. Comparing the KMeans cluster assignments with the true diabetes labels shows that the two clusters roughly correspond to the diabetic and non-diabetic groups.

6. Model Comparison & Evaluation

Overall Performance Summary:

Model	Accuracy	Precision (Diabetes)	Recall (Diabetes)	AUC
KNN	95.77%	88.65%	57.34%	0.9107
Decision Tree	97.02%	96.49%	67.28%	0.9712
Logistic Regression	95.90%	86.71%	60.86%	0.9561
Neural Network	96.96%	96.45%	66.51%	0.9721

6.1 Accuracy Comparison

All four models achieved accuracy above 95%. Decision Tree (97.02%) and Neural Network (96.96%) performed best, while Logistic Regression (95.90%) and KNN (95.77%) were slightly lower.

6.2 Precision & Recall Comparison

Precision measures how often a 'Diabetes' prediction is correct. Recall (Sensitivity) measures how many actual diabetes cases are correctly identified.

Model	Precision	Recall	Interpretation
KNN	88.65%	57.34%	Lowest recall, misses ~43% of cases.
Decision Tree	96.49%	67.28%	High precision, best balance with recall.
Logistic Regression	86.71%	60.86%	Lower precision and recall, linear boundary struggles.
Neural Network	96.45%	66.51%	Very similar to Decision Tree. High precision, moderate recall.

6.3 Confusion Matrices

Confusion matrices for all four models on the test set (7,728 samples: 7,074 No Diabetes, 654 Diabetes):

Model	True Negatives	False Positives	False Negatives	True Positives
KNN	7,026	48	279	375
Decision Tree	7,058	16	214	440
Logistic Regression	7,013	61	256	398
Neural Network	7,058	16	219	435

Key Observation: All models have very few False Positives, meaning they rarely misclassify a healthy patient as diabetic. However, False Negatives are significant, meaning many actual diabetic patients are missed. KNN has the highest number of False Negatives (279).

6.4 ROC Curves & AUC Scores

The ROC curve plots True Positive Rate vs False Positive Rate. AUC summarizes discriminative ability.

Model	AUC Score	Interpretation
KNN	0.9107	Good, but lowest of the group
Decision Tree	0.9712	Excellent discriminative ability
Logistic Regression	0.9561	Very good — strong baseline
Neural Network	0.9721	Best overall — highest AUC

The Neural Network achieves the highest AUC (0.9721), followed closely by Decision Tree (0.9712). KNN has the lowest AUC (0.9107) among the supervised models.

7. Conclusion

7.1 Understanding the Results

All supervised models achieved high accuracy (>95%) and excellent AUC scores (>0.91). The primary discriminating features — HbA1c_level and blood_glucose_level — are the strongest predictors. The Neural Network achieved the highest AUC (0.9721), demonstrating its ability to learn complex non-linear boundaries across 8 features simultaneously.

7.2 Model Performance Comments

- Neural Network (Best AUC: 0.9721): Learns non-linear decision boundaries effectively. Highest overall discriminative performance.
- Decision Tree (Best Accuracy: 0.9702): Interpretable and highly competitive. High precision (96.49%).
- Logistic Regression: A solid linear baseline. Lower precision and recall compared to tree/NN models.
- KNN: Achieved 95.77% accuracy but suffered the lowest recall (57.34%) and AUC (0.9107), showing distance-based classification is less effective here than decision boundaries.
- KMeans Clustering: Successfully identified two natural data clusters that roughly align with the diabetic/non-diabetic split.

7.3 Why These Results?

- High accuracy is driven by two dominant features: HbA1c_level and blood_glucose_level.
- Recall is moderate (~57–67%) across all models due to class imbalance. With 10.8x more negative samples, models learn a conservative boundary.

- The Neural Network and Decision Tree outperform Logistic Regression and KNN because the true decision boundary has non-linear components.

7.4 Challenges Faced

- High Missing Value Rate (~10% per column): Approximately 61,364 rows were removed during preprocessing.
- Severe Class Imbalance (10.8:1): The dominant class suppresses recall for the minority class.